Troubleshooting Partner with Artificial Intelligence (TPAI)

Levi Rodrigues da Silva*, Rodrigo Capobianco Guido†, Cleber Alexandre de Amorim‡ and Alexandre da Silva Simões§

*,§ *São Paulo State University (UNESP)*
Institute of Science and Technology of Sorocaba (ICTS)
Campus of Sorocaba, Brazil

†,‡ *São Paulo State University (UNESP)*
Av. Profa. Isette Corrêa Fontão, 505
São João da Boa Vista, SP, Brazil

Email: * *levi.silva@unesp.br,* † *guido@ieee.org,* ‡ *cleber.amorim@unesp.br,* § *alexandre.simoes@unesp.br*

***Abstract*—Industries in 2024 that produce tools, equipment, toys, accessories/housewares and consumer goods, all electronic, use the automation of a production line called SMT or Front End. The result of this production are generally components mounted on the Printed Circuit Board (PCB) forming electrical circuits for predetermined functions, called Printed Circuit Board Assembly (PCBA). These boards are tested to be mounted in your case, or chassis, generating the product for its intended purpose. This phase of production is known as Back End. All approved PCBA are then assembled into your consumer product and again tested in all their defined designations and with time and temperature stress, in an area that sometimes interconnects with the Back End, sometime called BOX. Failed PCBA are troubleshooted to recover the product and obtain lessons learned to improve the quality of the manufacturing process, reducing losses and scrap damage. The manufacturing industry has been working for with the possibility of eliminating some tests and troubleshooting in the Back End area, which doesn't add any value to the product. This is the purpose of this work, to apply Artificial Intelligence (AI) to check all data obtained from the Front and Back End, to guarantee only one sample test, preparing the industry to eliminate these unnecessary tests that don't add value to the product and create a troubleshooting tool that speeds up failure analysis and reduces scraps, [1] acting in this unwanted sector and as a result of this work, we support industries that are in the I3.0 revolution to updated to I4.0. [2]**

***Keywords*-Troubleshooting, Debug, Repair, Machine Learning, Artificial Neural Networks (ANN), SMT, PCBA, Industry 4.0.**

I. Introduction

The products were manufactured by hand for personal consumption or exchanged for other necessary products that weren't manufactured. When science invented steam engines and mechanization, widely used in the loom, producing in large quantities for distribution and sale to a much larger population, we have the beginning of industries in this revolution called industry 1.0 (1784). When electricity and combustion derived from petroleum arrived, a new impulse was triggered in these industries, modifying the production process on a scale, in an assembly line, which automated many workmanship jobs, no longer requiring a specific and qualified workforce. , bringing what we call industry 2.0 (1870) due to its great transformation. The evolution of electronics, robotic automation and the arrival of computers and the internet was another revolution in industries, called industry 3.0 (1960). With the arrival of Cyber Systems, the Internet of Things and the evolution of Artificial Intelligence (2012), we arrived at the industry 4.0 revolution. This latest revolution, still in development, leaves most companies stuck in what is called I3.0. The SMT lines of these electronic industries are what confers I3.0, being more than 90% of the products produced, also having a system that isn't possible to eliminate, due to components with terminals due to their size and large current consumption, using a process called PTH that represents less than 10% of production in 2024. This evolution of the industry is shown in figure 1.

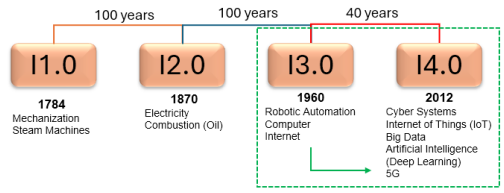


Figure 1. Evolution of the Industry

The important observation is that the first industrial revolutions took approximately 100 years to change and the last revolution that started in 2012 and is ongoing, took less than 40 years to adapt.

1) **Industry 3.0**, The I3.0 automated Printed Circuits Board Assembled (PCBA) in production in 2024 uses a production line called SMT that produces 90% of

PCBA and 10% of this production requires an older technology called Wave Solder, because in the line due the components Very heavy can cause failures until they reach the Reflow equipment and the type of solder used doesn't guarantee a high current at the component terminals if necessary. To solve these problems, another soldering method called Wave Solder is used and on PCBs, the terminals are normally used in metallized holes to support the weight and large currents. This process is called Pin Through Hole (PTH). Wave Solder presents small welding flaws that must be repaired and this manual process called Touch Up (TU) is part of this process, although big industries are eliminating this manual process to become completely automated. Therefore, 10% of products still go through the SMT process and also called the PTH process. This first part of using the bare board and placing all the project components, creating electrical circuits, is called Front End. After this PCBA is assembled, it goes through a process of separating each PCBA from its panel, as each panel can contain 1 board or hundreds of boards, depending on its size. Following the manufacturing process, the mother and daughter PCBA are tested, assembled in their final chassis or case (BOX), retesting all functions with stress and heat and finally sent for market consumption. This assembly and testing process before finalizing the product is called Back End. In all equipment in this automated process, 2 types of files are generated: One for operating setup and another with data on the manufactured products, however these data are currently available but with little or no use. One of the processes for testing motherboards and daughter boards, called Functional Testing (FT), industries want to eliminate or use as little as possible, as it doesn't add value to the product. Currently, this process is still necessary to reduce the number of faulty boards being sent for final assembly, as the losses would be greater. This entire system represented in figure 2 is what we currently call automated industry I3.0.

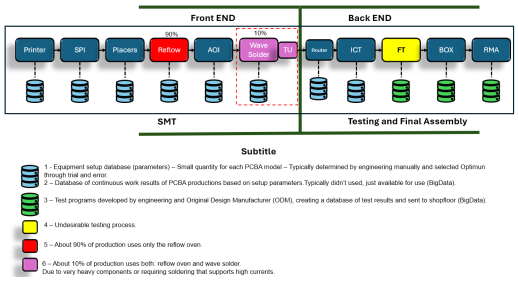


Figure 2. I3.0 Production Line

Explanation this infographic of the processes we have:

Printer: It has an exclusive accessory manufactured for the product, called a stencil screen, where the machine will automatically take the bare board and spread it on the islands and pads where the components will be assembled, with solder paste designed for that project and in exact quantity. All of this is programmed by software that indicates the area, quantity and exact height of solder paste for each of the components that will be assembled. This equipment is extensively studied and analyzed, as it is estimated that up to 60% of failures on PCBA arise from this equipment. There are other accessories such as the squeegee that will spread the solder paste and other materials mounted or used in this equipment.

SPI: After the PCB passes through the Printer and receives the harmoniously distributed solder paste, a test is carried out to check whether this process is within the parameters, using test equipment called SPI. This equipment also has a program with all the minimum and maximum solder paste data for each position on the board under ideal conditions. Through cameras, it can see in 3 dimensions and perform measurements of quantity, height, width and length, checking whether it is within the expected values and creating various measurement data that are stored in a dataset. Failed PCBs are given alarms and separated from other approved PCB.

Placer: All approved boards go through several Placer's, also known as Picks and Place, which are programmed equipment equipped with components in their feeders that were prepared in the setup. They also have cameras, sensors and several different tools connected by specific product type, to capture the component and place it in exactly the correct position on the PCB. Solder paste has a small adhesive in its composition, which holds the component exactly where it was placed. There are thousands of SMD components that are placed and depending on the complexity, several Placer's with small and big components such as BGA perform this task of assembling the components. Despite the great complexity, around 15% of failures occur in this sector.

Reflow: When the board is assembled, it passes through an oven with several temperature zones also programmed and prepared specifically for that product, in a template called Temperature Profile, and solders all the components on the board, thus completing the board. its cycle and exits as PCBA. This is an important step because it causes about 15% of failures.

AOI: Through conveyors, the PCB and PCBA are passed from equipment to equipment.The last of them in the SMT line is the AOI, an inspection equipment that, through cameras, programming with a golden or

standard model, compares the images of the PCBA that are passing through it, with its golden one, programmed with the minimum parameters and maximum, checking silkscreen readings of components, whether they are displaced, tombstone, short, inverted polarities, etc. If the quality parameters are tight, the test speed is slower, but if the speed is faster, certain faults may go unnoticed.

PTH: Practically more than 90% of current products are assembled in the SMT line, however 10% of boards require very heavy components that cannot be fixed to be soldered in the Reflow or their terminals require a large amount of current consumption, which makes the SMT type of soldering difficult, which can detach the terminal due to e heat generated.These components have pads where they are soldered, with metallized holes, and for this reason, they require a different Reflow oven to be soldered, called a Wave Oven or Wave Solder, in a very different process, which is usually summarized in these 3 letters PTH. Therefore, these boards pass the SMT process and also the PTH process. This is the last Front End process.

Router and ICT: At the beginning of the Back End process, the boards assembled into panels are separated (Router) and pass a test that is increasingly being allocated to the SMT process, called In Circuit Test (ICT). [3] This test examines as many of the components as possible in terms of their electrical function in any circuit, it also checks whether they have been soldered correctly using electromagnetic fields, whether the mounted LEDs are electrically ok, whether their colors are correct, whether there are no shorts on the board and the Sometimes even memory recordings are performed. These tests generate a detailed database of approved and failed boards. Its tests normally take from 30 seconds to 1.5 minutes and due to this quick response, its feedback on failures to correct SMT equipment and processes has a very large effect, which is why it adds value to the product and is increasingly adding value. to the SMT process.

Functional Test (FT): These are normally semi-automatic or even automatic equipment, which allow PCBA to be assembled and disassembled quickly, providing a quick test throughput, performing functional tests, that is, close to what the end user will perform, That's why the response to faulty boards takes a long time and contributes little to the SMT feedback. Only approved boards follow the assembly process or are stored in the Warehouse for later use. Failed boards are analyzed by a technical professional. After analysis, there are up to 3 attempts (strike) to replace a suspected faulty component. This component is replaced by a process called repair. If this performance doesn't work, the board normally goes to electronic waste, called Scrap. These functional tests don't add value to the product, because they will all be repeated in the next process when the product is ready to be consumed. It exists only due to human and process failures that cause many failures in PCBA and are introduced as a form of containment, as failures in the following process usually lose expensive parts, such as displays, chassis cases due to risks in the manufacturing process. assembly and disassembly to remove the PCBA for failure analysis.

BOX: Testing in this area is divided into 3 parts: The first of these is when the product is ready and fully assembled, carry out the tests as the end consumer would do. These tests involve 100% of the functional tests already carried out on PCBA, so the FT test doesn't add value, as it only checks the PCBA and not the product itself. The second test is stress, leaving the product in its main functions, running in repeated cycles for hours. The third test, sometimes sample, usually performs the same stress tests, but involving temperature variation, either well below the ambient temperature or much higher. A lot of data is also collected here and as the product is complete to be used in its packaging, this test is normally called BOX.

RMA: There is a sector sometimes called RMA, Return Material Authorization, which takes care of products that fail in the field and are under warranty. They return to the company or sector that provides services, to retrieve this information.

2) **Motivation for the TPAI Project** The my work carried out in the industries for 37 years, focused mainly on the area of PCBA troubleshooting and following several tools that reduced scraps, using the 6 sigma concept, but it wasn't enough to eliminate the unwanted the unwanted processand that causes a lot of financial loss. Knowing the SMT system how it works and witnessing the great search for industry 3.0 wanting to adapt and enter the new phase of this 4.0 revolution, pushed me to dream about this TPAI project. It started when I was part of a team of engineers from a big industry, which involved knowledge of front/end programming, Artificial Intelligence, troubleshooting and automation, in search of a solution to support troubleshooting technicians solve their work in less time and more efficiently This project didn't move forward, because the necessary professionals with more knowledge in Artificial Intelligence, with degree Master's and Doctoral degrees, went to other projects, but this contribution to this work helped a lot to form this concept of the **TPAI**.

3) **Unwanted process**, The flowchart in figure 3 shows where industries are focusing to reduce or eliminate

the process. There is the manufacturing process in the Front End and when it reaches the Back End, which is assembly and testing, the dream is that there will be no failures. As this isn't possible to be eliminated due to the complexity of the manufacturing process, You want there to be faults that can be quickly analyzed and feed back to the Front End, keeping this process real only during production.

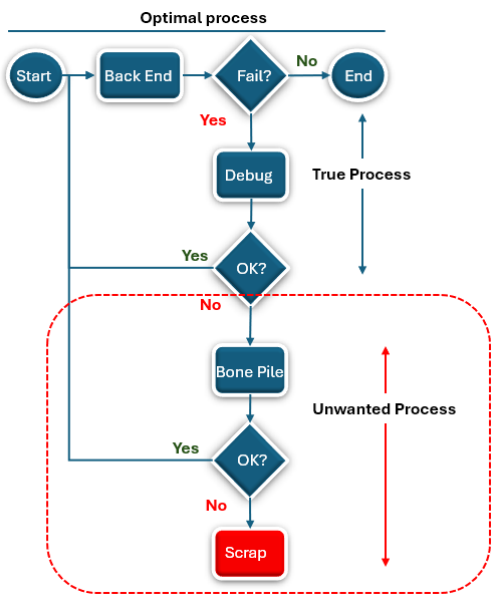


Figure 3. Unwanted process

The unwanted part is related to the Functional Test (FT) of mother and daughter boards, which makes it difficult to resolve failures during production. These boards are separated and sent to an area normally called the Repair Center or Bone Pile. There they create more financial losses with a long time stored failure analysis (Aging) and also when not carried out in 3 attempts (Strike), they are sent to scrap, causing more financial losses. The TPAI project intends to work in this area and for its success it requires the contribution of all other production phases, with valuable information during the manufacturing process. Therefore, we will have to use all the accumulated information data (Data Set) and the best tool for this work is to use Artificial Intelligence (AI).

4) **Other Resources** To use Artificial Intelligence and Data Sets, we need to capture this data on the equipment and then relate these Data Sets to each other and make some automated decisions. We will have to have a key called traceability, which is normally called the PCBA Serial Number, a sequence of unique identification numbers for that board. For this to happen, there will be many adaptations to SMT equipment that will involve the concept of Internet of Things (IoT) and when completed, this information will be able to be offered to the TPAI. This automation of the production line and contribution to the TPAI and other control systems will transform it from I3.0 to I4.0, being a side effect of my project that is highly desired by industries. To solve these problems and help with solutions, we will have to work with the Data Set before creating a system that involves hardware and software to perform failure analysis, which we call TPAI. Knowing the complexity of this process, we divided the project to be worked on in part to acquire the Master's degree and the second part to complete the project and obtain the Doctor's degree. This material will focus on the Data Set related to the first part of the project. This division of work will give time to look for the necessary partner: A manufacturing company (CM), plan the various phases of the process, learn the necessary tools and look for a team to complete this work when in the study phase to obtain the PHD title .

II. Artificial Intelligence

When we talk about Artificial Intelligence [4] we are entering a world that isn't so new, since there was the famous controversy by the Englishman Alan Turing, considered the father of computing, with the creation of the Turing Test: He defends the creation of a test hypothetical to analyze whether or not a computational system is intelligent like a human being, through the imitation test, using mathematics and thus giving the name to this hardware and mathematical concept that imitates a Perceptron or Adaline neuron. Since this concept was created, it has been used to solve human problems, using computing languages, the initial being LISP, later evolving into other languages and currently the best known and used is Phyton, where mathematicians and scientists in general use it to manipulate mathematical concepts and resolutions in search of an answer, imitating a human being in the way of thinking. Therefore, we will use Python and the concept of Machine Learning (ML), which means that the machine is capable of learning on its own, not depending on the action of programming. The AI is summarized below the example used ANN.

For the machine to learn, it can be supervised, that is, we give input data for the machine to learn, but we already know the result, to check whether the learning was good or bad, so we say that they are labeled. There are 2 types of learning: Classification and Regression. It can be unsupervised, where there are no labels or we don't know the final answer. It generally uses similarity or difference methods, which is why it has the following types of learning: Clustering, Association and Dimensionality Reduction. We have Reinforcement, where the only stimulus will be the

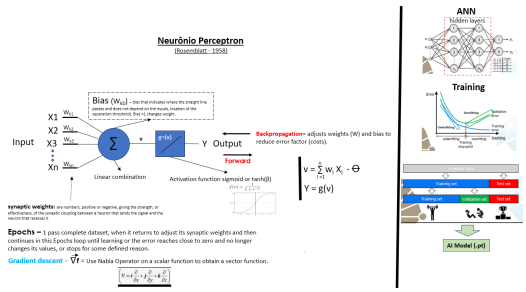


Figure 4. Enter Caption

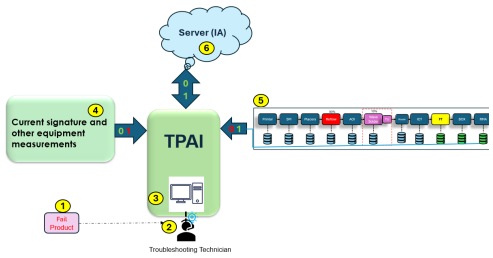
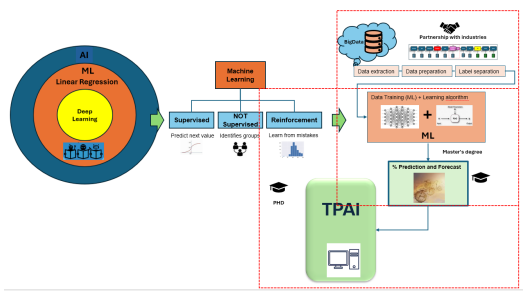


Figure 5. TPAI Project

reward. Unlike Unsupervised, which had no labels, but separated, for example, into groups. Here the machine must learn on its own, being prepared to receive rewards and penalties. This technique is very complex and, therefore, widely used in games, where there are rules that if followed there is a reward, life or additional weapons and if broken there is punishment, death, restarting the phase or loss of privileges. Figure 4 and 5 shows what Artificial Intelligence means and where I will use it in my TPAI project.



The concept of AI will involve finding a great partner in this work, where I intend to obtain the data and work on it. I must also define in all the databases which form of learning I will use, it may be that we will have to use all of them. There is the possibility of using Linear Regression for equipment Setup databases, as there is little data. Using Deep Learn on data from production machines that are huge. All this work will be carried out to achieve the Master's degree title. When we have configured and designed the various phases of construction of the TPAI, at the Doctor's degree, we will have to define the hardware (HW), the software (SW) and the firmware (FW) and we will have to return to AI to refine the various tools used and What will be the contribution of Back End and Front End to the project, already involving IoT and other concepts, therefore work that will involve other groups.

III. Proposed Approach

The **TPAI** project proposal figure 6 will be used for the explanation.

In the yellow circle with the number "1" we have the faulty board in Bone Pile and the following numbers follow the explanation. The troubleshooting technician makes his first analyzes in "2"and obtains new measurements not yet used in the testing process where the board failed, this in an HW built to communicate with the user and with the system where the AI is installed, whether in the cloud or on a local server, in "3 " and "4". All Datasets of the equipment in the manufacturing process have already been sent to the server, in "5". The server will run an AI model that contains all the information obtained from the 3 inputs: New measurements in "4", production data "5" and useful information from the troubleshooting technician in "2". The result of this analysis will be a prediction of which component position is at fault. As this prediction may be incorrect, the troubleshooting technician will also pass on all the information about the board in its manufacturing process, using the Serial Number traceability key. On the computer display with all this information, the technician will be able to follow the AI prediction or make another more accurate decision with all the data in hand. Therefore, the partnership of AI and human intelligence acting together forms the name of this TAI project.

IV. Materials and Methods

We still don't have the ideal partner for this project and we have already started inviting two big CM companies: First installed in the city of Sorocaba and second installed in the state of Parana. To analyze a database and its complexity, we had access to 3 databases, which were used as an example of working with AI. A dataset from the beginning of the SMT process called SPI.csv where it stores the manufacturing data of a product on the SPI equipment that analyzes the performance of the Printer equipment. We also have two files in the final phase of the testing process, one called V123.csv and the other Board.csv. We use the Google tool and notebook called Colab to access and open these files. The first dataset analyzed was V123.csv with 3748 record lines and using Phyton we checked this structure. This dataset consisted of boards analyzed and troubleshooted, but we found that despite the majority being approved in a new

retest, we had boards with scrap, others continued to fail and a number analyzed are in the process of exchanging components to be retested (repair). These repair boards can add up to approved, failed boards or go to scrap. We focus on analyzing each column in this file to reach a conclusion. This dataset was from the BOX test area where there were many PCBA failures but also process failures where they were inserted. Next we get the dataset called Board.csv. which is a product of the unwanted test area, called Functional Test (FT). We also check this database in its architecture main. In this case, we separated the 3 main failures and called other failures those that were inferior. We had to adapt some data that had the same meaning but was written differently. We used Binary Regression in this database, to find out whether the AI successfully separated the signs with fail from the approved ones. Here we had an obstacle, the data was strings and we had to adapt this data to be worked with Binary Regression using the correct commands for transform string into binary. With logistical regression used and separating 80% for the AI to learn and 20% for testing, we arrived at an excellent result. Then we went to the SPI.csv database where the data was already adequate and was easier to work with in Logistic Regression. We also work with the Artificial Neural Networks (ANN) to check the practicality of this tool.

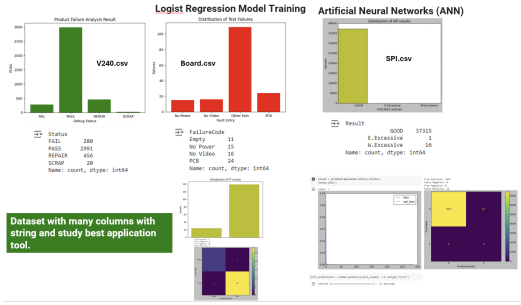


Figure 6. Linear Regression and ANN in DataSet

All the analysis steps mentioned are available in the TPAI.ipynb file because we didn't write these commands in this document, as they are just for testing purposes and would take up the space we need to detail the TPAI project. [5] This work isn't final, it was just an exercise to learn about the difficulties and ease of working with AI tools, in this case Linear Regression and ANN, also to have the first contacts with the files and start practical learning of ML. The Supervised Machine Learning (SML) discipline presented many tools to be studied and applied to the most complex datasets we will come into contact with their. Another tool used to write this document was to access latex through OverLeaf.

V. Results and Discussion

To start the research work, we adopted which **database** we would use and decided that there will be 5: ScienceDirect, Scopus, Scielo, Web of Science and Google Scholar. We also adopted one of the **methodologies** offered by SageResearchMethods and the Brazilian ABNT standards. The searches continue and so far we haven't been able to find any work similar to what we are proposing. We will have to refine this search before **guaranteeing** that this work is unique.

Figure 7. Data Base and Methods

As a student at UNESP we can have access to any journal or scientific research material, being able to use the information with this tool made available to us and research our information and then even publish it in a database to contribute to other scientists in search of information.

VI. Conclusions

This work leads to many conclusions, some of which we mention:

- Complex work of relating a failure to all databases in the manufacturing process.
- Achieve a partnership to obtain real databases (Big Data and Setup) and apply the desired technique.
- Reducing scraps, aging in the Bone Pile and troubleshooting with 1 Strike will be the beginning of eliminating the unwanted process.
- Challenge of having SerialNumbers as a traceability key in all equipment.
- Use Linear Regression and Deep Learning. The challenge is to learn all the techniques and know how to apply them correctly to databases.
- To have equipment that works together with a concept of obtaining BigData, AI analysis, troubleshooting and decision processes, called TPAI, it will be necessary to divide it into master's and doctoral theses.

Acknowledgements

We would like to thank our supervisor for this project, Prof. Dr. Rodrigo Capobianco Guido and the professors of the Scientific and Technological Research Methodology

disciplines Prof. Dr. Cleber Alexandre de Amorim and Supervised Machine Learning (SML) Supervised Machine Learning (SML) Prof. Dr. Alexandre da Silva Simões for the great learning offered.

To UNESP for the opportunity for this development and coexistence with science.

REFERENCES

[1] L. Molina-Fernández, D. Arias-Aranda, and O. Bustiza-Sánchez, "Industry 4.0 and competitive advantage. an analysis from the perspective of knowledge management — industria 4.0 y ventaja competitiva. un análisis desde la gestión del conocimiento," *Dyna (Spain)*, vol. 99, pp. 150–153, 2024.

[2] B. P. Santos, A. Alberto, T. D. F. M. Lima, and F. M. B. Charrua-Santos, "IndÚstria 4.0: Desafios e oportunidades," *Revista Produção e Desenvolvimento*, pp. 111–124, 2018. [Online]. Available: http://revistas.cefet-rj.br/index.php/producaoedesenvolvimento

[3] M. Ivanova and N. Petkov, "Machine learning for in-circuit testing of printed circuit board assembly." ACM, 12 2021, pp. 221–228.

[4] S. Erduran and O. Levrini, "The impact of artificial intelligence on scientific practices: an emergent area of research for science education," *International Journal of Science Education*, pp. 1–8, 2 2024.

[5] Levi Rodrigues da Silva, "Project tpai dataset: Linear regression and ann," Drive Google Colab: https://drive.google.com/drive/folders/15Fu30TdlB-X5R4J-iELpPAZp_bbLUkcQ?usp=sharing. Sorocaba, june 2024.